

Learning When to Look Like vs. Work Like: A Dual-Expert Attention with a Graded Alpha Knob

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Abstract - The Short Story

Standard attention only asks one question: 'do q and k look similar?' – that's $E0 = q \cdot k / \sqrt{d}$. It works for 'dog' vs 'dogs', but I kept failing on 'Paris' → 'France' or '5+3' → '8'. They don't look similar, they work together. So I split attention into two experts in my notebook. $E0$ is 'looks-like' and $E1 = -(1/d) ||W_{rel} q - k||^2$ is 'works-like' – a lock-and-key, does my key W_{rel} turn q into k? Then I needed a knob α to decide which expert to trust. I learned the hard way α is NOT a fixed 0.2 or 0.9. Those numbers I used before were just examples to explain. Real α is adaptive, continuous in $[0,1]$, evaluated from $E0$ strength by a frozen library L (DeBERTa-MNLI). If L sees no relation → $\alpha \approx 0$ (pure similarity). If weak relation like car→road → α small, around 0.18-0.25 (I used to write ~ 0.2 as shorthand, 80% similarity + 20% relation). If strong relation like car→wheel or king→queen → α large, around 0.87-0.93 (I wrote ~ 0.9 , 10% similarity + 90% relation). I never use $\alpha=1.0$ – in March my retrieval collapsed, 'queen' started retrieving 'queen bee', so I keep 10% similarity as anchor. The rest of the paper is how I predict $\hat{\alpha}$ and why I needed a decision boundary β – I wasted 3 days in v2.7 because target and distractor kept flip-flopping when I nudged α .

1. Why I needed two experts

I started with what everyone knows:

$E0(q,k) = q \cdot k / \sqrt{d_{model}}$ (1) [in v2.5 I removed $\sqrt{d_{model}}$ and gradients exploded – lesson learned]

$E0$ alone failed on relational queries. '5+3' should attend to '8' but dot-product is tiny. So I added second expert:

$E1(q,k) = -(1/d_{model}) ||W_{rel} q - k||^2$ (2) [I tried $q^T W k$ in v2.4 – symmetric, didn't work for directed Paris→France]

Best is 0 when $W_{rel} q = k$. Note scales differ – $E0$ unbounded, $E1 \leq 0$. I was confused for a week why $E1$ always dominated, until I realized I need LayerNorm per query across keys:

$\tilde{E0}(q,k) = (E0 - \mu_q)/\sigma_q$, $\tilde{E1}(q,k) = (E1 - \mu_q)/\sigma_q$ (2b) μ_q, σ_q over keys for fixed q – I used to call it d vs d_b and got mixed, now I call dimension d_model

2. Alpha - the mixer knob that is NOT fixed

Oracle mixing I use:

$S^*(q,k; \alpha) = (1-\alpha) \tilde{E0} + \alpha \tilde{E1}$ (3) α is DJ mixer knob

Here is where I was unclear before, so let me fix it once and for all – α is NOT 0.2 or 0.9 fixed. I wrote those as examples in my notes to explain 80/20 vs 10/90, but reviewers thought they were discrete labels. They are not.

Real definition I use now:

$\alpha(q,k) = L(q,k) = \sigma(\Phi_{rel}(q,k) - \gamma_{nli} \cdot \text{Sim}(q,k)) \in [0,1]$ continuous (4) $\gamma_{nli}=0.3$

Φ_{rel} is entailment logit from frozen DeBERTa-v3-large MNLI – premise = text(q), hypothesis = text(k). $\text{Sim} = \cosine(q,k)$ which proxies $E0$ strength. γ_{nli} (I used to just call it γ and then reused γ for annealing – that was my bad, now I split to γ_{nli} and γ_{mix}). L is frozen, not dependent on trainable W_{rel} to avoid circularity – otherwise model cheats.

If Φ_{rel} says no entailment → $\alpha \rightarrow 0$. If weak 0.3-0.6 → α in $[0.15, 0.30]$ – I shorthand it as ~ 0.2 , like car→road. If strong > 0.8 → α in $[0.80, 0.95]$ – I shorthand ~ 0.9 , like car→wheel. These are density peaks from isotonic regression on my 2k human labels ($\kappa=0.78$), not hard bins. α is evaluated from $E0$ strength via L for each pair.

I keep saying this because I was told ' α is fixed' – it is not. It is graded, adaptive, continuous. Library L evaluates E_0 strength and gives me α .

3. Predicting Alpha - my first attempt failed

I first tried only q :

$$\hat{\alpha} = \sigma(W_{\theta} q + b_{\theta}) \quad (5) \text{ – collapsed to mean, useless}$$

Then I fixed it to pairwise – needs both q, k and both experts:

$$\hat{\alpha}(q, k) = \sigma(W_{\text{gate}} [q; k; E_0; E_1] + c_{\text{gate}}) \quad (6) \quad W_{\text{gate}} \in \mathbb{R}^{1 \times (2d_{\text{model}} + 2)} \text{ – clipped to } [-10, 10] \text{ else gradient } \hat{\alpha}(1 - \hat{\alpha}) \rightarrow 0$$

$$\hat{S}(q, k) = (1 - \hat{\alpha}) \hat{E}_0 + \hat{\alpha} \hat{E}_1 \quad (7)$$

4. Decision boundary - the flip-flop that wasted 3 days

In v2.7 I saw target t and distractor p swapping winner when I nudged α a little. I thought my code was buggy for 3 days. Then I plotted score vs α' – it's just two lines crossing.

$$S^*_t(\alpha') = (1 - \alpha') \hat{E}_{0,t} + \alpha' \hat{E}_{1,t}, \quad S^*_p(\alpha') = (1 - \alpha') \hat{E}_{0,p} + \alpha' \hat{E}_{1,p} \quad (8)$$

$$\text{Flip where } S^*_t = S^*_p \quad (9) \rightarrow$$

$$\beta_{\{t,p\}} = (\hat{E}_{0,p} - \hat{E}_{0,t}) / [(\hat{E}_{1,t} - \hat{E}_{0,t}) - (\hat{E}_{1,p} - \hat{E}_{0,p}) + \varepsilon \cdot \text{sign}(D)] \quad (10) \quad \varepsilon = 1e-6, D = \text{numerator diff} \text{ – early version I divided by zero and got NaNs for an epoch}$$

If $|D| < \varepsilon$ lines parallel $\rightarrow \beta = \text{None}$. If $\beta \notin [0, 1]$ target dominates always $\rightarrow$ skip. β is photo-finish where two horses cross as you turn knob. It's independent of oracle α – it depends only on $\hat{E}_0, \hat{E}_1$.

5. Weighting near the cliff - spotlight

If oracle α is near β , small error flips winner. I care more there:

$$d_b = \sqrt{(\alpha - \text{sg}[\beta])^2 + 1e-4} \quad (11) \text{ distance to boundary, sg=stop-gradient – I used to let } W_{\text{rel}} \text{ move } \beta \text{ away to cheat, now I stop it}$$

$$w_{\{t,p\}} = \text{sg}[1 + \lambda_s \exp(-d_b / \tau_s)] \quad (12) \quad \lambda_s = 2.0, \tau_s = 0.1 \text{ – spotlight near cliff}$$

$$s_{\beta} = \tanh((\alpha - \text{sg}[\beta]) / 0.05) \quad (13)$$

$$L_{\text{decision}} = \text{softplus}(-s_{\beta} (\hat{\alpha} - \text{sg}[\beta]) / \tau_d) \quad (14) \quad \tau_d = 0.1 \text{ – tried hinge in v2.7, exploded, softplus worked}$$

If α left of β , $\hat{\alpha}$ must also be left, else penalty. w says shine more light near cliff.

6. Final loss and the annealing trick

$$L_{\{v2.8\}} = w_{\{t,p\}} (\hat{\alpha} - \alpha)^2 + \lambda_d L_{\text{decision}} \quad (15) \quad \lambda_d = 0.5$$

I had 8% drop from exposure bias when training only on oracle α . Fixed with mixing:

$S_{\text{mix}} = (1 - \alpha_{\text{mix}}) \hat{E}_0 + \alpha_{\text{mix}} \hat{E}_1$, $\alpha_{\text{mix}} = (1 - \gamma_{\text{mix}}) \hat{\alpha} + \gamma_{\text{mix}} \alpha$, γ_{mix} annealed $1 \rightarrow 0$ over 10k steps – early uses stable oracle, later matches inference $\hat{\alpha}$. W_{rel} gets grads from task CE on S_{mix} , NOT from w or β thanks to $\text{sg}[]$. W_{gate} gets grads only from $L_{\{v2.8\}}$.

7. What it feels like

$\alpha = 0$ pure similarity – 'dog' and 'dogs' look alike. α small ~ 0.2 (0.18-0.25) is cousin who looks like you but is your accountant – car $\rightarrow$ road, 80% looks 20% works. α large ~ 0.9 (0.87-0.93) is king $\rightarrow$ queen – 10% looks 90% relation. β is where two horses cross. w is spotlight. L_{decision} says don't flip winner near photo-finish.

8. How to reproduce - my setup

$d_{\text{model}} = 512$, heads=8, AdamW lr $3e-4$ for W_{rel} (orth init), $1e-3$ for W_{gate} (Xavier), c_{gate} init $\sigma^{-1}(\text{mean } \alpha) \approx -0.5$, grad clip 1.0, batch 128.

$D_{\alpha} = 12k$ pairs: bAbI 16-18 4k, CLUTRR 4k, ConceptNet 4k (IsA, PartOf, Capitals). Negative 1:1 random keys. Split 10k/1k/1k. Human 2k subset for calibration $\kappa = 0.78$. $p(\alpha)$ is continuous – peaks at 0 (45% mass in

[0,0.05] no relation), ~ 0.2 (25% mass in [0.15,0.35] weak), ~ 0.9 (30% mass in [0.75,0.95] strong) – after isotonic regression, not discrete labels. That was my confusion before – I wrote 0,0.2,0.9 and people thought discrete.

Oracle: $\Phi_{\text{rel}} = \text{DeBERTa-v3-large MNLl logit, premise}=\text{text}(q), \text{ hypothesis}=\text{text}(k).$

$\text{Sim}=\text{cosine}(\text{all-MiniLM}). \gamma_{\text{nli}}=0.3$ subtracts similarity to avoid rewarding look-alike. Thresholds calibrated via isotonic regression.

I wrote this in Mosul, v2.8.3 – trying to keep it clear for researchers but still my voice. α is graded, adaptive, evaluated from E0 strength via L, not fixed. 0.2 and 0.9 were just my shorthand for weak and strong ranges. – Abdulrahman Al-Hayali – May 2026